

THE CURRENT STATE OF PEOPLE ANALYTICS

A practitioner's perspective.

TABLE OF CONTENTS

1

2

3

4

5

6

Overview

Organization

Technology

Skills

Vendors

Resources

What is people analytics?

The case for people analytics

Increasing demand

People analytics position within the firm

People analytics within HR

Technology options
Technology needn't be expensive

What a people analytics team needs
Demand for People analysts

Job descriptions for the new breed

Market segmentation for vendors

Supporting information

About Organization View

OVERVIEW

What is People analytics and why has it become one of the hottest areas of HR?

WHAT IS PEOPLE ANALYTICS?

For a firm there are three economic reasons to measure and analyse data:

- *because we can use this data to make more effective decisions*
- *because by measuring and presenting data we can change behaviours in a way that is of value to us*
- *because we can sell the data.*

People analytics is an evidence-based approach for making more objective and effective business decisions where the workforce is a major driver of performance.

It is at once both new and old. Old in that businesses have for a long time wanted to use information and statistical skills to better understand and manage the productivity of the workforce. New in that a combination of factors, mostly we think external to HR, have brought it the top of the HR agenda.

Technology has changed for ever how businesses collect, store and deal with data. The popular phrase 'Big Data' was originally coined to describe the technology challenges created by vast volumes, often of unstructured data that are being generated by firms every second. It has come to mean, in the popular mind at least, the ability to spot previously undiscovered patterns in data that can be used to create value for the firm, their customers, shareholders, communities and employees.

People analytics mostly focuses on better decision making. We'll cover the types of questions people are answering through data in this document.

What has driven this renewed activity? We see a few underlying factors:

- we have more data than ever before, and the quality is better
- there has been a massive drop in the cost of technology needed to store and analyse the data
- business as a whole has become more analytical. This has shifted managers' expectations
- there has been an increasing number of articles in the specialist and general press describing how the leading firms are using people analytics

THE CASE FOR PEOPLE ANALYTICS

Leading Analytic Organizations have 17% better talent outcomes than other organizations

Corporate Executive Board

High performing organizations have higher levels of sophistication in their use of analytics and are more likely to have adopted analytics across all aspects of human capital management

EY, CFO - CHRO report

Quantified organizations have 79% higher Return on Equity than non-quantified Organizations

Sierra-Cedar 2014 Survey

Only 15% of respondents use margin / profitability as their most important measure of HR effectiveness

Accenture research

People costs are often nearly 70% of all operating costs of a firm, and for many firm the major creator of value, but historically the analysis and optimisation of the workforce has either been missed, or done purely from a cost perspective.

At the moment, only 16% of organizations have people analytics capability in HR¹. Adoption is highest in the Retail / Wholesale sectors. Only 9% of HR departments are capable of predictive analytics.

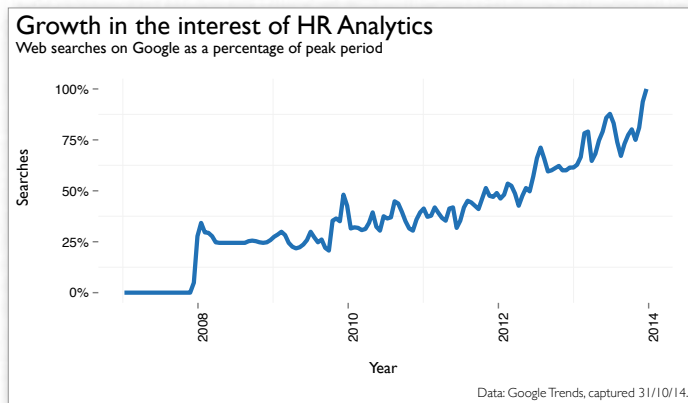
This is reflected in decision making. According to the CEB, less than one quarter of business leaders use data from HR for key talent activities. According to the same report just 17% of business leaders believe that talent analytics are focused on the right issues.

Leading firms like ABN AMRO drive business performance from people analytics. ABN AMRO have helped business leaders understand how talent decisions effect customer satisfaction, quality and products sold.

INCREASING DEMAND

14% of companies have analytics capability in the HR function

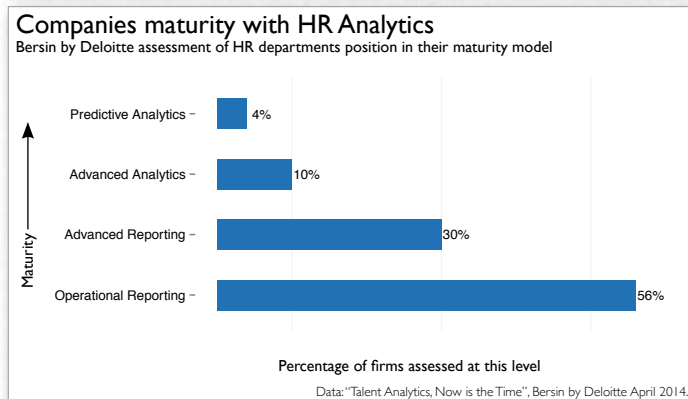
57% of HR teams increased their investment in measurement & analytics in 2013¹



Analytics is arguably one of the hottest areas of interest for HR. Interest via web searches has increased 4-fold over the last 4-5 years. It is, however starting from a relatively low base with only a small number of firms doing the more advanced, and high value types of analysis.

Value in analytics is typically achieved at higher levels of maturity. Decisions need to be made based on predictions of what is likely to happen therefore workforce measures need to be provided as forecasts or predictions - forward looking - rather than historic measures.

As a comparison to the Bersin assessment of HR Analytics maturity (shown to the left), Accenture's 2012 survey² of business analytics practitioners reported that 51% were conducting predictive analysis. We believe this gap puts pressure on HR to raise its game.



Where is the demand typically coming from?

The executive team (CEO, CFO, COO) demand it. Most of the strong use of analytics that we see is being driven by the executive team.

Investors are demanding it. Reporting is changing. Initiatives such as the Global Reporting Initiative are encouraging firms to communicate internal capabilities, which include workforce measures. Corporate Social Responsibility reporting frequently includes workforce criteria.

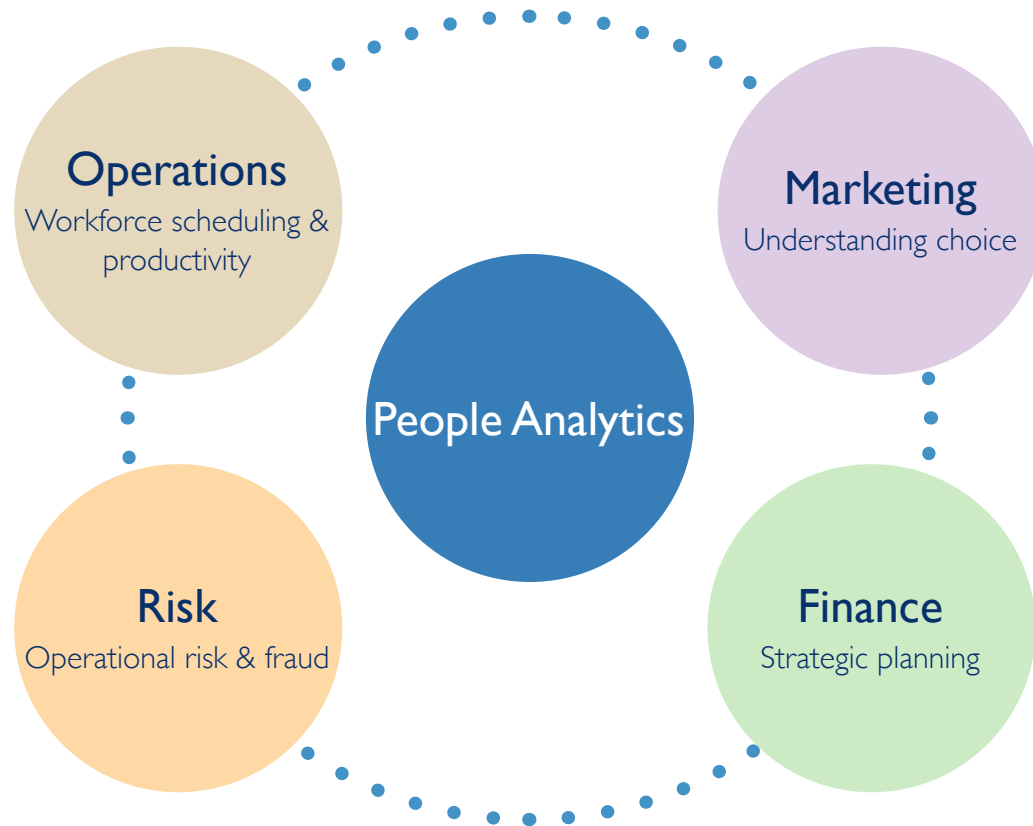
Regulators are demanding it. In some sectors regulators are showing an interest in workforce related measures.

ORGANIZATION

How firms are structuring their people analytics capabilities

PEOPLE ANALYTICS POSITION WITHIN THE FIRM

People analytics functions need to collaborate closely with other functions to be successful. In some instances it is sitting outside HR.



For too many firms, HR analysis of workforce data has been too introspective - measuring and reporting the things that HR does, mostly for the benefit of HR. That is changing.

We're seeing an active interest in analysing HR data from other parts of the firm. As Simon Constance of EY noted:

"I've worked with two clients with the [Analytics Centre of Excellence] reporting into operations, with a dotted line to HR. This can help with demand insight and getting into the productivity agenda."

Several of the accounting firms have published research in the last two years of the increasing collaboration between the CHRO and CFO.

It is quite clear that the appetite for good workforce analysis across firms is strong. We see a risk that if HR doesn't meet the demand others will & HR departments will lose a key aspect of workforce strategy.

PEOPLE ANALYTICS WITHIN HR

The people analytics centre of excellence

People analytics Centres of Excellence are rapidly becoming the accepted way of deploying analytics capability within HR departments.

In some ways this mirrors the way other HR specialist groups are being deployed in the post-Ulrich model. However, even compared to other specialist functions the people analytics team are likely to have widely different backgrounds than the rest of HR.

The trend matches a business-wide approach to deploying analytics. There are considerable benefits to be realised from centralising analytics, concentrating talent, developing resilient processes, coordinating data governance, deploying specialist technology and ensuring re-use of work.

Within HR the CoE should also be the greatest advocate of data-driven-decision making, evangelising an approach rooted in reality.

Its position within HR

The CHRO will likely be the person most likely to be seeing the demand from their peers in other functions for people analytics, and this, combined with the CoEs contribution to developing and monitoring effective workforce strategies means that in most instances we're seeing the people analytics CoE reporting directly to the CHRO.

There is another aspect which suggests that this is the right approach. If done well, people analytics is likely to uncover evidence that will challenge some of the approaches and policies HR has been promoting. In many instances we find that traditional accepted truths in HR can't be justified by the available data.

This can call for a degree of independence for the CoE from their peers in HR. We see a parallel to how an internal audit function typically reports to the Chairman's office.

TECHNOLOGY

How are firms building a platform to do this work?

TECHNOLOGY OPTIONS

Technology won't solve the people analytics problem, but an early integrated plan can reduce problems later.

In their 2014 survey of HR Technology, research firm Sierra-Cedar describe 4 different approaches to delivering HR Analytics technology:

- **Embedded HR:** analytics as part of HR technology
- **Standalone HR Analytics Tool:** e.g. Visier
- **Enterprise BI:** reporting tools such as Business Objects or Tableau or analytics tools such as SAS or R
- **Combinational Solutions:** using a mixture depending on task.

Our experience matches Sierra-Cedar's :

Once an organization gets a taste of analytics, it will want to expand and incorporate data from other sources which will necessitate the adoption of an enterprise BI platform solution.

Businesses will increasingly be getting embedded HR analysis tools as part of their HR systems, especially when they are cloud-based. Experience of early adopters though suggests this might increase the appetite to implement enterprise BI solutions. The benefit for implementing such technology will be primarily cost-reduction; moving reporting that is predominately manually produced to automated systems.

Technology is providing significant opportunities to do analysis that was previously uneconomical. However the technology shouldn't be allowed to drive what analysis is done. Far too many firms fall into the trap of analysing what is easy to analyse, not what has a higher potential value to the firm.

Enterprise BI, especially the analytic tools such as R and SAS require skilled analysts. They also require that effort is spent on data preparation and integration, often with data from outside HR. As Sierra-Cedar notes, for most big firms the answer will be combinational solutions. The challenge will be to get the different parts of the portfolio to work together.

TECHNOLOGY NEEDN'T BE EXPENSIVE

Many of the technologies used for people analytics will be open-source. The trade-off for lower technology costs is needing higher skilled analysts.

Many of the firms at the cutting edge of people analytics are using open-source technology. In KDNuggets annual survey of data mining technology, seven out of the top ten most used technologies were open source.

We, like Google's People Analytics team, use the open-source statistical programming language 'R'. Its background is in academia, with many statistical papers and textbooks now having their own R packages - extensions to the tool set. We even use R packages that were developed by the People Analytics team at Google.

As well as R and several other prominent statistical tools, many of the data storage tools - the so called 'NoSQL' databases that underpin many 'Big Data' firms will be open-source. Without licence costs the time and ease of prototyping dramatically drops. Data analytics teams should be agile.

Doing analysis in code provides three particularly important benefits:

- Code makes data analysis auditable - it's much easier to explain where data came from if every step of the analysis is written in a script
- Once a piece of code is written it can be re-run every time new data is available
- Code can often be reused in new projects, dramatically reducing the time / cost to do new projects.

Most statisticians leaving universities today will have coding skills and the ability to process large quantities of messy data. However, our experience of traditional HR reporting teams, especially those historically using tools such as Excel is that they're unlikely to have the necessary skills.

With a combinational technology approach these lower-skilled analysts will be replaced, probably with fewer but higher-skilled analysts

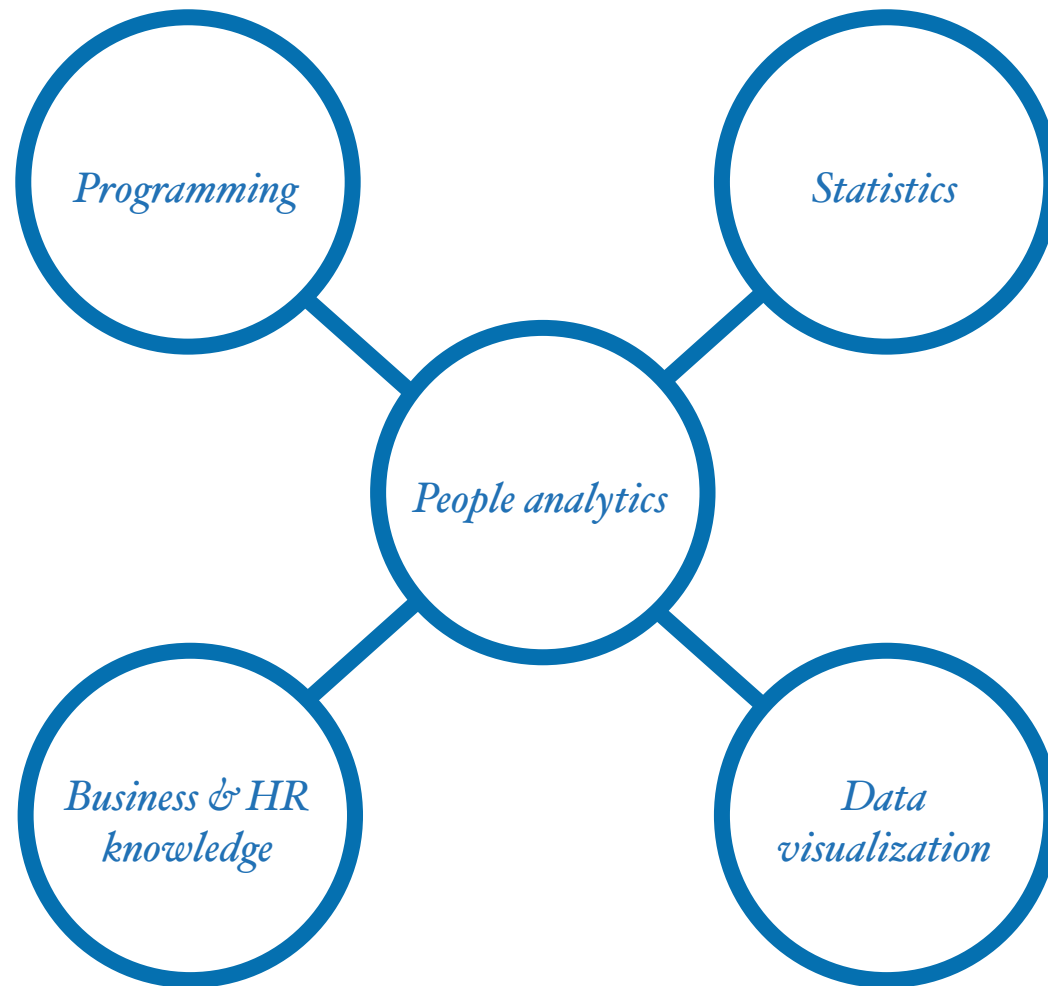
SKILLS

People analytics requires new skills for HR

4

WHAT A PEOPLE ANALYTICS TEAM NEEDS

Analytics teams require different skills than are common in most HR departments. The challenge is how to build such multidisciplinary teams.



Good analyst teams require a multi-disciplinary skill set. It's highly unlikely that analysts will have all skills themselves.

The most important skill is domain expertise - deep knowledge of HR and Business. Knowing what questions to ask, what research is available and models of workforce development is essential to realising value from analytics.

Statistics knowledge is vital for knowing how to interpret data. Increasingly statistical techniques will include technology-driven techniques such as machine learning and simulation. Many are hiring analysts with advanced degrees to meet these needs.

The most important part of communicating data is the communication, not the data. Data visualization skills are vital for ensuring the correct story is read from the data, and for encouraging people to change.

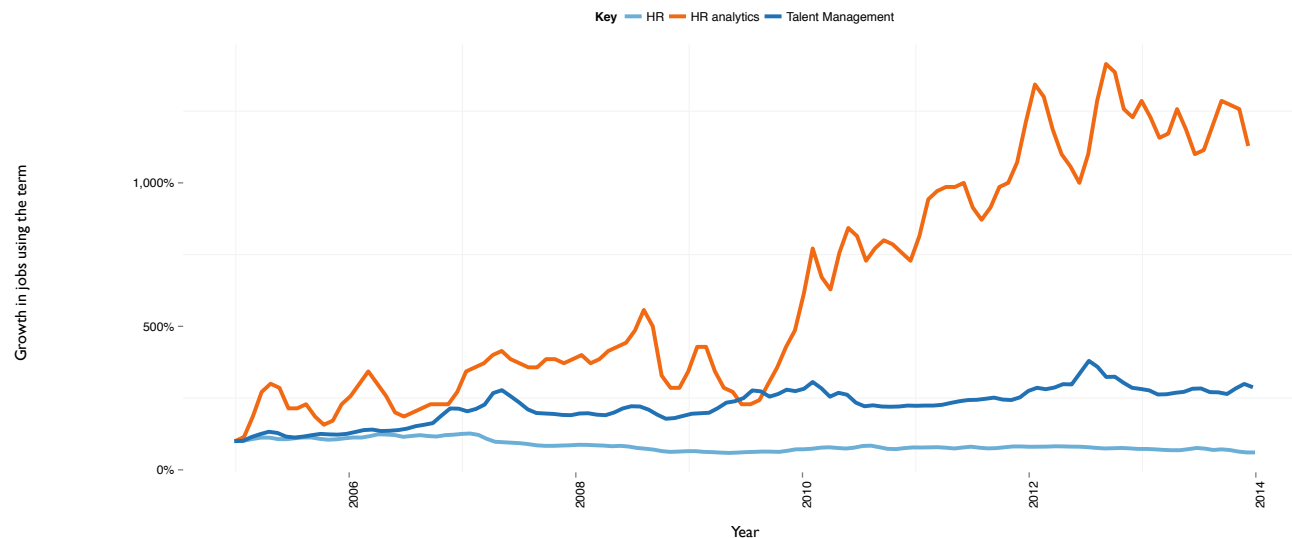
Finally, as data gets bigger and more complex programming skills come to the fore.

DEMAND FOR PEOPLE ANALYSTS

Analytics teams require different skills than are common in most HR departments. The challenge is how to build such multidisciplinary teams.

Growth in the jobs advertised for HR analytics

Occurrence of the following terms in job advertisements. Values were originally percentage of total jobs and were changed to index values based Jan 2005



Data: Indeed.com job trends, captured 31/10/14. Data visualisation by OrganizationView

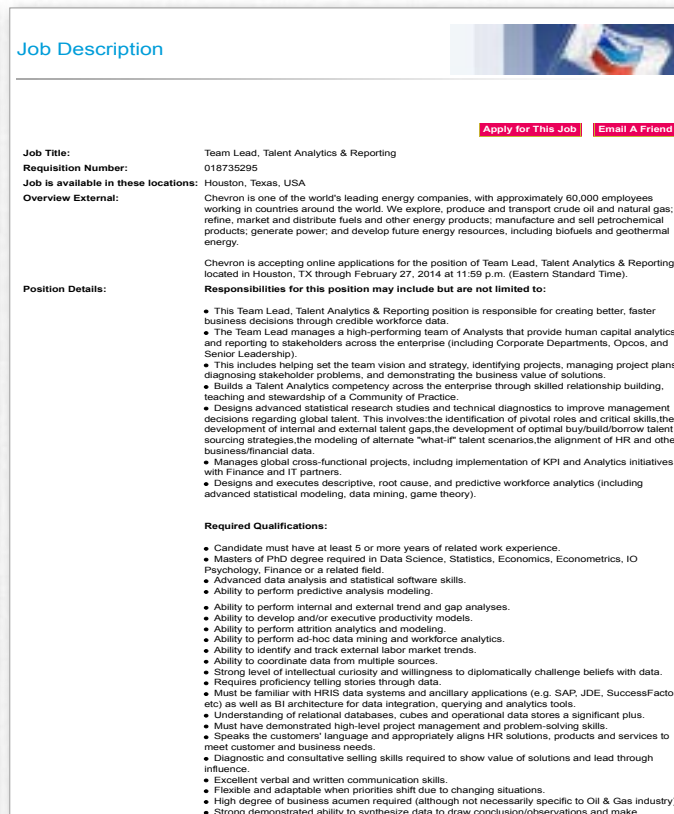
The challenge for HR will be how to recruit these skills. McKinsey suggested that the US alone faces a shortage of 140-190,000 people with analytical expertise.

HR needs to recruit these individuals, which are in some instances significantly different from the traditional HR background in face of competition from a large number of other functions and businesses. It also needs to articulate what careers analysts can expect in HR.

Salaries are significantly higher. Careerbuilder reported that Statisticians and Economists in the UK will earn over twice as much as HR administrative occupations, rising to 3x in the top firms.

JOB DESCRIPTIONS FOR THE NEW BREED

Analysis of job descriptions is a good way of seeing how the market is developing



HR has previously hired a reasonable number of psychologists. Whilst this is likely to continue, only a subset of psychologists are likely to have the technology skills required for this work.

The number of job descriptions we see asking for Phd level statistics, economics or even science subjects like physics is growing. Whilst some of these jobs have unrealistic, or even probably unnecessary requirements the trend is clear - these are data scientists not traditional HR analysts.

Several firms are rotating data scientists from other functions through HR. This can work well as long as there are members of the team with strong HR knowledge. Marketing, Operations Research, Finance and Economics teams could be good internal sources.

VENDORS

Where to go for support: how the vendor market is shaping up.

MARKET POSITIONING FOR VENDORS

Suppliers can be segmented into five groups based on their dominant business model or market offering. Many suppliers will have some activity in secondary groups as well as their main position.

The Technologists

The Technologists take an IT-centric approach to analytics, with a revenue model that is heavily based on software licences and implementation services. In today's market software is likely to be cloud-based.

A Technologists approach is likely to be successful if you are convinced that your requirements for analytics are predictable. Alternatively they may be useful to cover the basics in a hybrid model.

The Data Aggregators

There are several firms whose strength, and to some degree value, is from their large stores of cross-firm workforce data, usually at the level of an individual employee. These firms are now creating new business lines through the aggregation and analysis of this data.

During the last few years several large acquisitions have been linked to this approach. These would include CEB's \$660m acquisition of SHL and IBM's \$1.3bn acquisition of Kenexa. Previously conservative firms like ADP are worth watching in this space.

The Business / HR Consultancies

Most, if not all of the major business consultancies with HR teams are offering People Analytics service lines. Whilst many have built, and in many instances retired, People Analytics technology, their project-based services are likely to meet the needs of many

HR buyers. EY is a good example of a firm in this space. Assignments are likely to be in the areas of workforce planning, workforce productivity & performance improvement modelling and setting up and advising analytics CoEs

The Start-ups

Some of the most exciting and advanced analytics currently being done in HR falls within this sector. Most firms in this sector make one or two products targeting niche areas. Many of these are in the area of hiring, especially for certain in-demand groups such as

IT skills. The availability of large volumes of public data - including social media data - make machine learning to identify patterns possible. These services can be great add-ons to your analytics approach

Analytics as a Service Providers

Analytics as a Service firms can provide an analytics CoE capability on an outsourced basis, either to manage fluctuating capacity or specific skills. Most firms in this category will bring their own technology which enables them to automate common activities,

delivering economies of scale. Whilst in other functions or sectors the market is established it is new within HR. The relatively small size of deals has made HR unattractive to date for the larger firms such as Accenture, Mu Sigma and Fractal.

RESOURCES

Where to learn more.

6

SUPPORTING INFORMATION

The following people provided advice and assistance during the creation of this document:

Simon Constance

Director, HR advisory, EY

Patrick Coolen

Manager, HR metrics & Analytics, ABN AMRO

Stacey Harris

VP Research & Analytics, Sierra-Cedar

For further information on HR Analytics the following are good sources:

Analytics in Action: Breakthroughs and Barriers on the Journey to ROI
[Accenture](#)

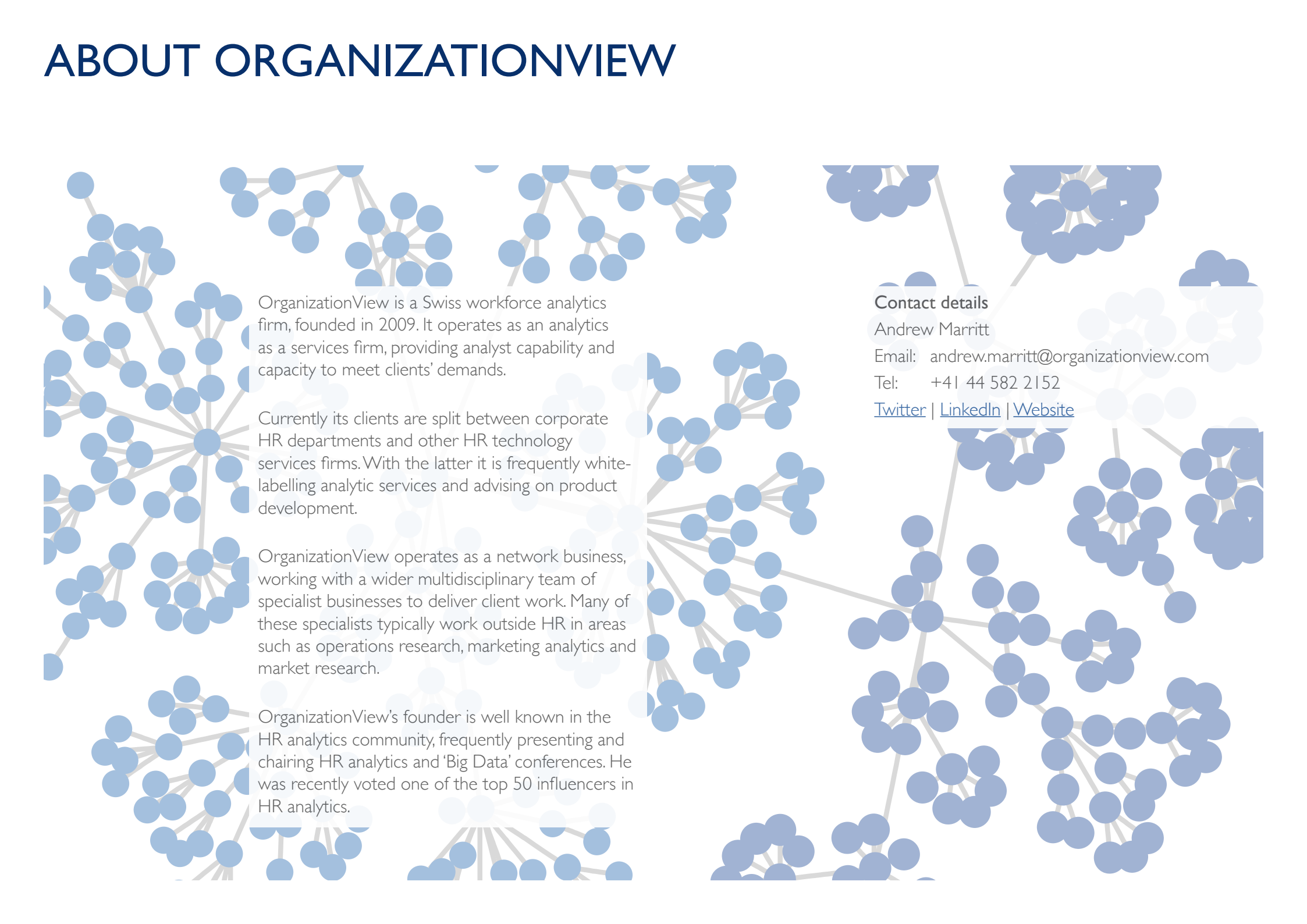
Workforce Analytics: Numbers behind HR
[Deloitte](#)

Question for your HR Chief: Are we using our 'people data' to create value?
[McKinsey Quarterly](#)

17th Annual HR Systems Survey
[Sierra-Cedar](#)

The Analytics Era: Transforming HR's Impact on the Business
CEB Corporate Leadership Council (membership required)

ABOUT ORGANIZATIONVIEW



OrganizationView is a Swiss workforce analytics firm, founded in 2009. It operates as an analytics as a services firm, providing analyst capability and capacity to meet clients' demands.

Currently its clients are split between corporate HR departments and other HR technology services firms. With the latter it is frequently white-labelling analytic services and advising on product development.

OrganizationView operates as a network business, working with a wider multidisciplinary team of specialist businesses to deliver client work. Many of these specialists typically work outside HR in areas such as operations research, marketing analytics and market research.

OrganizationView's founder is well known in the HR analytics community, frequently presenting and chairing HR analytics and 'Big Data' conferences. He was recently voted one of the top 50 influencers in HR analytics.

Contact details

Andrew Marritt

Email: andrew.marritt@organizationview.com

Tel: +41 44 582 2152

[Twitter](#) | [LinkedIn](#) | [Website](#)